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Multi-Modal Deep Learning in Breast Cancer Diagnosis: A Review of Recent Advances

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***Abstract*—Breast cancer detection often requires multiple imaging modalities to overcome the shortcomings of each modality. Deep learning advancements facilitated our incorporation of these multi-modalities into precise diagnostic systems. This paper reviews the recent applications of multi-modal deep learning in breast cancer detection and classification. This approach is further consolidated through recent advances such as multi-view vision transformers and different fusion strategies utilized in networks to leverage complementary information between modalities, improving both tumor detection and lesion characterization. These works represent a key advance toward developing AI systems that can replicate radiologists' comprehensive reviews without additional burden on healthcare ecosystem. We also addressed clinical barriers in implementation, such as data heterogeneity and interoperability, and suggested pathways for future directions to overcome these challenges. This study describes the steps required for a safe and effective integration of multi-modal deep learning into healthcare facilities in near future.**

***Index Terms*—multimodality, deep learning, breast cancer diagnosis, multiview**

I. Introduction

Breast cancer remains one of the most significant global health concerns and is the leading malignancy among women, contributing to a substantial proportion of cancer-related mortality [1]. Based on its progression, breast cancer is classified into two main categories: *in situ* and invasive. The invasive type is further divided into four stages, reflecting the extent of cancer spread, which is an essential criterion for determining survival rates [2]. Early diagnosis plays a crucial role in reducing the need for invasive treatments, improving survival rates, and ultimately enhancing patient outcomes.

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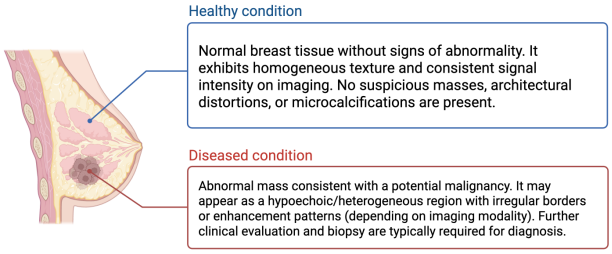

Fig. 1. Breast cancer appears as irregular texture of breast tissues.

Medical imaging techniques provide invaluable insights for breast cancer diagnosis and treatment planning. However, in many cases, the detection of abnormal tissues in breast is challenging due to the similarity of normal and abnormal structures (Fig. 1) in standard imaging modalities such as mammography. In real-world scenarios, the need to utilize multiple imaging modalities for decision-making is well recognized. For example, although mammography is the standard imaging technique for breast cancer screening, it has limitations in distinguishing cancerous tissue from normal tissue in dense breasts [3]. This highlights the necessity of more comprehensive diagnostic tools that can enhance accuracy and improve patient care.

Deep learning models are increasingly used to assist radiologists by automating image analysis and reducing diagnostic workload. While traditional approaches typically rely on a single imaging modality, recent advancements integrate multiple modalities to combine complementary information from different sources. This advancement brings deep learning models closer to the current clinical workflow, enhancing their practical applicability. The diverse information sources used in deep models can stem from different imaging modalities, such as combining mammography and magnetic resonance

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imaging (MRI), or from different sequences within the same modality, such as T1- and T2-weighted MRI scans. Additionally, multiple views from a single modality, such as the mediolateral oblique (MLO) and craniocaudal (CC) views in mammography, can also be utilized to improve diagnostic accuracy.

This paper reviews recent advances in multi-modality deep learning for breast cancer diagnosis. It starts by outlining key imaging modalities used in detection and provides an overview of common deep learning architecture components. The next section covers strategies for integrating multi-modal data, highlighting their strengths and limitations. A discussion of existing studies follows, focusing on how each approach processes and integrates data. The paper concludes with an exploration of current challenges, future directions, and key insights.

II. Breast cancer imaging modalities

A variety of imaging modalities play crucial roles in the detection, diagnosis, and management of breast cancer. Each technique offers unique advantages and limitations, contributing different types of information about breast tissue and potential abnormalities.

A. Mammography

Mammography uses low-dose X-rays to capture 2D images of the breast and is a widely used tool for breast cancer screening [4]. Standard exams include two views per breast: the MLO, showing the posterior and axillary regions, and the CC, offering a top-down view. Combining both views improves diagnostic accuracy, especially for detecting abnormalities such as masses and microcalcifications, early signs of cancer.

However, mammography has limitations, particularly in dense breasts where low-dose X-rays reduce image clarity. Its 2D nature can also cause information loss due to overlapping tissue. Digital Breast Tomosynthesis (DBT), or 3D mammography, addresses this by capturing multiple angled images and reconstructing high-resolution 3D views, improving cancer detection and reducing false positives [5]. Moreover, mammography is known as a pain-full imaging modality that cause a remarkable discomfortable during acquisition.

B. Ultrasound

Ultrasound (US), or sonography, is a non-invasive imaging method that uses high-frequency sound waves to visualize breast tissue without ionizing radiation. It is especially effective in detecting small, node-negative cancers obscured by dense tissue in mammography. A handheld transducer scans the skin, assessing lesion vascularization and vessel patterns [6], [7].

However, hand-held breast ultrasound (HHBUS) faces limitations, including operator dependency, lack of standardization, limited field of view, and time-consuming procedures. Accurate interpretation requires significant expertise to distinguish between benign and malignant changes, anatomical structures, and artifacts [6].

C. Magnetic Resonance Imaging (MRI)

Magnetic Resonance Imaging (MRI) uses strong magnetic fields and radio waves to produce high-resolution 3D images of the breast without ionizing radiation [6]. Dynamic Contrast-Enhanced MRI (DCE-MRI) highlights abnormal vascularity using gadolinium-based contrast, while Diffusion-Weighted Imaging (DWI) assesses water molecule diffusion, often restricted in malignant tissues, without contrast [8].

MRI is valuable for high-risk patients and imaging dense breasts where other methods may fall short. However, it has limitations, including low specificity that may lead to false positives and higher costs. Advances in DWI are helping improve its diagnostic value and accessibility [8].

D. Positron Emission Tomography (PET)

Positron Emission Tomography (PET) is a functional imaging technique that detects metabolic activity in breast tissue using a radiotracer, typically 18F-FDG, which accumulates in cancer cells due to their high glucose metabolism. PET is valuable for staging, assessing lymph node involvement, detecting metastases, and monitoring treatment response. It is often combined with CT (PET/CT) to add anatomical detail and improve diagnostic accuracy [9].

Despite its sensitivity in detecting active tumors and suitability for whole-body imaging, PET is not used for routine screening due to limited spatial resolution, high cost, and reduced efficacy in identifying small or low-metabolism tumors. The need for radiotracer injection and radiation exposure also restricts repeated use [9].

III. Multi-Modality Deep Learning Approaches

Several multi-modality deep learning models have been proposed for breast cancer diagnosis, leveraging different imaging modalities for tasks such as classification and segmentation. In this section, we first discuss the most widely used deep learning architectures for multimodal breast cancer diagnosis, including Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and Graph Neural Networks (GNNs), see Figure 2 for an illustration between different paradigms for a classification task. Subsequently, we examine the primary strategies for fusing information across different modalities and then review the latest advances in this field for various diagnostic tasks.

A. Deep learning architectures

Convolutional Neural Networks (CNNs) are deep learning models effective in processing imaging modalities like mammography, MRI, and ultrasound by extracting relevant features from images [10]. Their layered architecture captures spatial patterns through convolution and pooling, followed by fully connected layers for tasks like classification. While CNNs excel at learning local features, they struggle with capturing global context. To overcome this, newer architectures like Vision transformers and advanced CNN variants have been developed to enhance the receptive field and improve feature representation.

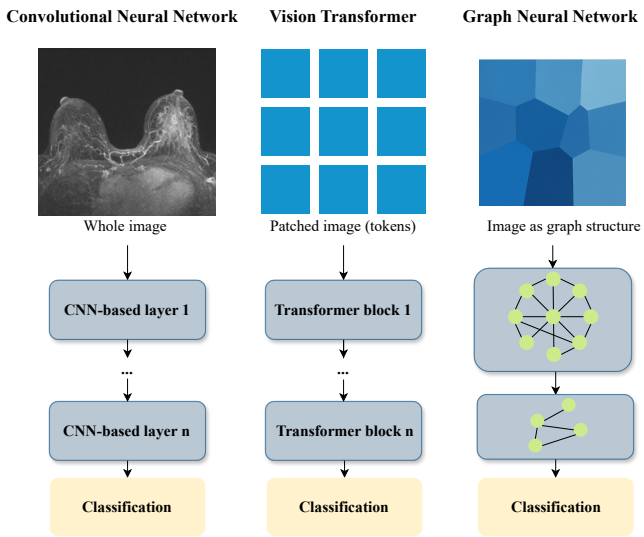

Fig. 2. Comparison of deep learning architectures for image understanding. Convolutional Neural Networks (CNNs) process the entire image using hierarchical feature extraction through convolutional filters. Vision Transformers (ViTs) divide the image into fixed-size patches and apply self-attention mechanisms to model global context. Graph Neural Networks (GNNs) represent the image as a graph of nodes connected by edges, capturing structural and relational information across the image.

Vision Transformers (ViTs) have emerged as powerful models for capturing long-range dependencies in image data using self-attention mechanisms [11]. By dividing images into patches, ViTs and hybrid CNN-transformer models offer a more holistic understanding of features. Approaches like the Swin Transformer [12] enhance this by applying attention within sliding windows, expanding the receptive field over layers. These architectures are especially promising for breast cancer diagnosis, particularly when large annotated datasets are available, as transformers generally require substantial data for effective training.

Graph Neural Networks (GNNs) provide a powerful framework for modeling structured relationships in medical imaging, particularly in scenarios where spatial and topological information is crucial [13]. While CNNs and transformer-based models process image data in a grid-like manner, GNNs utilize graph-structured representations, enabling them to capture complex interactions between different anatomical regions or multimodal data sources. By incorporating GNNs, deep learning models can align more closely with clinical decision-making processes and facilitate imaging analysis beyond the pixel level.

B. Multi-modality strategies

Multimodal deep learning can be categorized into several approaches, including fusion, representation, alignment, translation, and co-learning, each designed for specific applications with distinct underlying assumptions and limitations. Among these, fusion techniques are the most widely used for disease diagnosis, as they enable the integration of information from multiple modalities at different stages of a deep model [14]. Figure 3 illustrates different fusion approaches for multi-modality processing. **Early fusion**, also known as data-level fusion, involves combining raw data or low-level features from different modalities into a single representation before being fed into the deep model. This approach preserves the most detailed information from all modalities and effectively captures cross-modality dependencies. However, a major limitation of early fusion is its requirement for perfectly aligned data, making integration into clinical workflows challenging due to the need for additional preprocessing steps for some tasks. Moreover, it is difficult to scale for additional modalities and is highly sensitive to missing data.

Intermediate/joint fusion, also known as feature-level fusion, assumes that features extracted from different modalities contain complementary information. In joint fusion, features obtained from separate modality-specific models are combined and fed into another deep model to enhance the learning process. This approach adds flexibility by allowing the use of attention mechanisms and dynamic weighting, making it more adaptable. Additionally, it can handle misaligned data more effectively compared to early fusion. However, it is important to note that suboptimal feature extraction in one modality can negatively impact overall performance. Therefore, designing task-specific models tailored to each modality is a critical aspect of the process.

Late fusion, also known as decision-level fusion, combines predictions from separate models trained on different modalities at the decision stage. The fusion of predictions is typically performed using techniques such as weighting, majority voting, or averaging. The key assumption behind late fusion is that individual modalities provide independently meaningful information, while the aggregated decisions leverage the complementary strengths of each modality. This method offers the highest robustness against noise in any single modality and scales well when incorporating additional modalities. However, it overlooks fine-grained interactions between modalities and tends to be suboptimal compared to early and joint fusion when the modalities exhibit strong interdependencies. Different combinations of these techniques can also be employed, which is known as hybrid fusion.

C. Multimodal advances for breast cancer diagnosis

Several studies have employed late fusion strategy for multi-modal breast cancer diagnosis. One approach utilized deep neural networks and CNNs to process gene expression and copy number variation data alongside pathological images, with the predictions fused through weighted aggregation. The fusion weights were optimized using Simulated Annealing to improve subtype classification accuracy [15].

Ensemble-based approaches have also demonstrated strong potential for enhancing multi-modal integration in breast cancer diagnosis. In one study, CNNs were utilized for feature extraction from clinical, gene expression, and CNA data, and a stacked ensemble model was employed using classic machine learning classifiers. Random forest outperformed the other classifiers, attributed to its ability to aggregate multiple

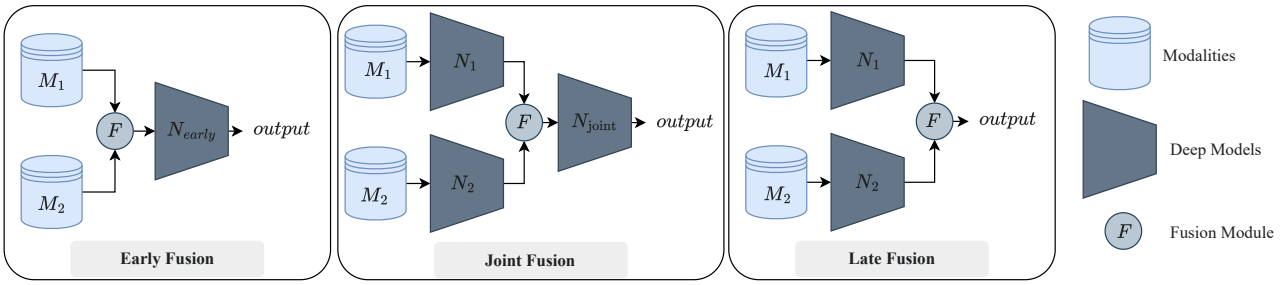

Fig. 3. Different approaches for multi-modality fusion in deep models.

decision trees and handle imbalance of the training data effectively [16]. Another study combined six patient scores derived from clinical and MRI data using multiple model variations initialized with different random seeds, with the final decision based on the average of all model scores [17]. In [18], traditional radiomic and deep learning features from ultrasound, mammography, and MRI images were combined using a weighted voting strategy. Additionally, stacked generalization with XGBoost as a meta-classifier was explored to integrate modality-specific predictions and produce the final diagnosis.

In [19], authors fused output probabilities from a feed-forward network trained on clinical data and a ResNet50 model fine-tuned for DCE-MRI projections, feeding them into a small neural network for final prediction, enabling decision-level integration without requiring cross-modal feature interaction. A recent framework integrated mammography and histopathology images using two CNNs: mammoCNN, designed for single-channel mammography images, and histoCNN, tailored for three-channel histopathology images. The extracted features were optimized using a binary optimizer (BEOSA), which reduced dimensionality by eliminating non-essential features. These optimized features were then passed to a softmax classifier to generate probability distributions for each modality. The outputs from both networks were scaled to a uniform label set and concatenated, forming a combined probability map. This fusion layer enabled comprehensive classification of breast cancer abnormalities, with the final result derived from the optimized probability map [20].

Late fusion plays a crucial role in processing heterogeneous data, such as combining imaging with genomic or clinical information. For instance, Hussain et al. [21] integrated imaging features extracted from a modified SE-ResNet50 with clinical features processed via an artificial neural network. These separately learned representations were concatenated and passed through a classification layer, enabling the model to leverage complementary information for enhanced diagnostic performance. [22] developed a hybrid deep learning model that integrates features from histopathological images and gene expression profiles, extracted using CNN and MLP architectures, respectively. The features were fused at the decision level to accurately classify breast cancer molecular subtypes. Another approach used a late fusion framework where modality-specific CNN classifiers were trained independently, with their outputs combined using a Dempster-Shafer Theory (DST)-based aggregation scheme. This scheme incorporated each model's predictive uncertainty, quantified via positive predictive values on the validation set, achieving superior performance compared to standard weighted averaging across all modality combinations [23]. In [24], a multimodal architecture was proposed that processes mammography, ultrasound, and clinical metadata. Imaging features were extracted using ResNet-18, while clinical data were encoded via a multi-layer perceptron. To enable inter-modal learning, cross-attention transformer blocks allowed mammography and ultrasound features to enrich one another, followed by self-attention to aggregate them into a unified imaging representation. This was then fused with clinical metadata using late fusion and passed through fully connected layers for both T1–T5 classification and a binary benign versus malignant decision.

Some studies have employed feature-level data integration within deep learning models to enhance feature representation and improve classification performance. For instance, in [25], the authors proposed a model for benign/malignant classification using both ultrasound and mammography images. Features were initially extracted using CNN-based layers from each modality and statistically analyzed for significance. The selected features were then fused and passed through an LSTM-based model for final classification. A key limitation of this approach was its dependence on manual selection of regions of interest (ROIs) during preprocessing, which requires expert intervention and hinders scalability. In a related study, [26] developed a deep model for predicting the four molecular subtypes of breast cancer using ultrasound along with CC and MLO mammogram views. They introduced both inter-self-attention and intra-self-attention mechanisms to effectively model correlations within modalities. Additionally, the incorporation of channel and spatial attention modules further enhanced the model's ability to capture rich and relevant features, improving its predictive performance.

Another study [27] proposed a model for classifying breast MRI images using two modalities: diffusion-weighted imaging (DWI) and apparent diffusion coefficient (ADC) maps. The authors designed a series of attention modules to exploit the correlation, complementary, and discriminative information between modalities. Initially, features were extracted using

CNN-based layers. An inter-modality attention module based on feature correlation was then applied to capture relationships across modalities. This was followed by a channel-attention module designed to integrate modality-specific features at the channel level, along with an auxiliary loss function to guide the model toward more task-relevant representations. The resulting features were passed through a dual-parallel module consisting of spatial and channel attention mechanisms. Finally, an additional correlation-based attention module was applied to further refine the representations, ensuring that the most informative and modality-relevant features were emphasized for accurate classification. In [28], the authors utilized T1 and contrast-enhanced T1-weighted (T1c) MRI sequences for breast cancer segmentation. Their approach involved an initial localization model that performed a coarse pre-segmentation to roughly identify the tumor region. The pre-segmentation output was then concatenated with each modality separately and fed into two parallel branches. Multi-stage fusion modules were subsequently applied to integrate the features from both branches and progressively refine the segmentation output, resulting in more accurate and detailed tumor delineation.

Several studies have leveraged multi-view information as a form of multi-modality adaptation to enhance the performance of deep learning models. Most of these works focus on multi-view analysis of mammography images, rather than processing the views independently or using simple fusion techniques, where crucial inter-view correlations may be lost.

In [29], the authors introduced a cross-view attention module based on transformers to address the misalignment issues between different mammography views. Instead of combining local features—which can be adversely affected by spatial misalignment—they aggregated information at the global feature level. This strategy led to superior performance compared to single-modality processing and late fusion methods. [30] extended their approach to four-view mammography analysis with the baseline of Swin transformer instead of ResNet. The main idea for the proposed cross-view attention was utilized in other studies as well [31]. Similarly, in [32], the authors proposed a shifted-window-based dynamic attention block to fuse information from the two standard mammography views: MLO and CC. By incorporating both self-attention and cross-attention mechanisms, their method effectively captured intra-view and inter-view relationships. The proposed approach outperformed the baseline Swin transformer in classifying breast cancer as benign or malignant. In [33], a two-view deep learning model was proposed, inspired by state-of-the-art object detection frameworks. The approach in this study emphasizes following morphological cues—similar to the diagnostic strategy employed by radiologists—rather than relying on geometrical cues, which are often misaligned across different mammographic views.

While some studies focus on two-view analysis, both left and right breast views enhances diagnostic performance by providing richer anatomical context. In [34], the authors utilized all four standard mammography views—RCC, RMLO, LCC, and LMLO. Each view was divided into patches and mapped into a latent space. After adding positional embeddings, the embedding vectors were processed through a local transformer block with shared parameters to capture intra-view dependencies. The outputs were then concatenated and passed through a global transformer to model inter-view dependencies. Finally, an additional MLP layer produced the classification results, demonstrating the superiority of their approach on a private dataset. In [35], the authors utilized a pre-trained multi-view model for breast cancer classification. To ensure consistency in orientation, the left-side mammography images were horizontally flipped to match the orientation of the right-side images. Feature extraction was performed using a ResNet-22 backbone, after which features from the MLO and CC views were concatenated and passed through two fully connected layers. The resulting features from the left and right sides were then aggregated to produce the final prediction. The model was further fine-tuned using transfer learning on a new, unseen private dataset, demonstrating its adaptability across different data distributions.

Graph Neural Networks (GNNs) have also been employed as a core architecture for mammography mass detection by leveraging multi-view information. In [36], an anatomy-aware graph convolutional network (AGN) was introduced to facilitate multi-view reasoning. The AGN framework involves three main stages: first, a graph convolutional network is used to model the semantic and geometric relationships between the CC and MLO views. Second, another graph-based model captures the similarities between bilateral views (i.e., the same view from the left and right breasts). Finally, a correspondence reasoning enhancement module is applied to the constructed graphs to fully exploit the multi-view correlations. Graph-based models have also shown promise in cancer subtype classification using multi-omics data treated as multi-view inputs. For instance, in [37], a graph convolutional network was employed to extract features from various omics data sources, while attention mechanisms were used to integrate the multi-omics information effectively.

Multi-view deep learning models have also been applied to other imaging modalities, such as ultrasound. In [38], the authors proposed a dual-branch deep model to process ultrasound slices extracted from the transverse and longitudinal planes. The outputs from each branch were combined using a late aggregation strategy. To further enhance performance, they introduced two attention modules: a region-guided module (RGM) for spatial attention and a time-guided module (TGM) for temporal attention.

IV. Challenges and Future Directions

Despite progress, deploying multimodal deep learning for breast cancer diagnosis faces significant hurdles. A major issue is the lack of comprehensive, well-aligned datasets, as hospitals often collect different modalities at varying times and scales. Missing data, alignment difficulties, especially with 3D or serial imaging, and class imbalance further complicate training. Privacy regulations also limit data sharing, hindering

dataset development. Architectural and computational complexity is another barrier. Fusing high dimensional modalities can create large, over-parameterized models prone to overfitting, especially with small datasets. Fusion strategies remain task and modality dependent, with adaptive weighting still under research. Transformer-based models, while powerful, require large datasets and are computationally intensive.

Inference speed and model robustness also pose challenges. Efficient design, such as lightweight backbones and modality-specific feature selection, is needed to ensure reliability in real-world settings, including when data quality varies. Training with data ablation can improve robustness. Finally, interpretability and fairness are critical. Multimodal models often act as "black boxes," making it difficult for clinicians to trust their decisions. Explainability methods, like attention maps, are advancing but still maturing. Bias is another concern, as models can reflect demographic imbalances in training data. Addressing fairness requires diverse datasets and techniques like adversarial training and fairness aware loss functions to ensure equitable outcomes and reduce healthcare disparities.

Advancements in multimodal diagnostic systems depend on several key areas. First, large, well-annotated multimodal datasets built from pooled hospital and research resources are essential for training robust models. Handling missing modalities through imputation or flexible architectures is also critical, enabling models to function with partial inputs. Second, novel fusion methods are emerging. Graph-based models treat each modality as a node with learned relationships while evolving transformer-based systems aim for unified cross-modal attention across images, text, and more. As computational power grows, models could incorporate full patient records, offering a comprehensive diagnostic view. Integrating domain knowledge such as physics for imaging or biology for omics can enhance model generalization, clinical relevance, and data efficiency. Third, explainability, clinical validation, and integration are vital. Models should provide interpretable insights from each modality and be tested through user interface studies, external validations, and prospective trials. Transparency in data usage and collaboration with regulators are needed for clinical adoption. If these challenges are addressed, multimodal deep learning could become a powerful clinical tool for identifying missed cancers, tailoring diagnoses, and ultimately improving early detection and patient outcomes.

V. Conclusion

Multimodal deep learning models are proving highly effective for breast cancer detection, outperforming single-modality approaches by combining data from mammography, ultrasound, MRI, and more. Models that incorporate cross-modality and cross-view relationships enhance lesion detection and classification, especially in complex cases. However, challenges persist, including the lack of diverse, annotated datasets, limited interpretability, and difficulty handling missing or imperfect modalities. Regulatory and workflow integration issues also require collaboration across clinical, technical, and policy domains. Despite these hurdles, multimodal deep learning shows strong potential to advance precision breast cancer care through holistic and validated approaches.

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